Lecture 8 (*March 5*)

Reading: Section 3.3
Viewing: Lecture 8

We will now put together the ideas of the previous two lectures in order to find the complete (a.k.a. general) solution $\boldsymbol{v}$ of the system:

$$A\boldsymbol{v} = \boldsymbol{b} \tag{63}$$

for any $m \times n$ matrix A. First of all, as we have seen in Fact 5, there are no solutions if the $m \times 1$ vector $\boldsymbol{b}$ is not in the column space $C(A)$. So we henceforth assume that $\boldsymbol{b} \in C(A)$, which is just saying that there exists a particular solution:

$$A\boldsymbol{v}_{\text{particular}} = \boldsymbol{b} \tag{64}$$

This $n \times 1$ vector $\boldsymbol{v}_{\text{particular}}$ is <u>a solution</u> of (63), but it may not be the only one. In fact, for any other solution $\boldsymbol{v}$ of the original system (63), we may subtract the two equations above and obtain:

$$A(\boldsymbol{v} - \boldsymbol{v}_{\text{particular}}) = 0$$

So the difference $\boldsymbol{v} - \boldsymbol{v}_{\text{particular}}$ lies in the nullspace of A. This immediately leads to the following.

Fact 6. *The general solution of the system* (63) *is given by:*

$$\boxed{\boldsymbol{v}_{\text{general}} = \boldsymbol{v}_{\text{particular}} + \boldsymbol{w}_{\text{general}}} \tag{65}$$

where:

- $\boldsymbol{v}_{\text{particular}}$ *is a particular solution of* (63)*, which exists if and only if* $\boldsymbol{b} \in C(A)$
- $\boldsymbol{w}_{\text{general}}$ *is any element of* $N(A)$*, i.e. the general solution to* $A\boldsymbol{w}_{\text{general}} = 0$

The fact above is a general mathematical feature of <u>linear equations</u>. For example, it also holds in the theory of linear ordinary differential equations, as you can see in 18.03.

Reduced row echelon forms allow you to deal with both bullets in Fact 6 at the same time. Let me show you how to do this in an example. Consider the system:

$$\begin{bmatrix} 2 & 6 & 2 & 2 & -1 \\ 2 & 6 & 1 & -1 & 2 \\ 3 & 9 & -1 & -9 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -7 \\ 6 \\ -6 \end{bmatrix} \tag{66}$$

Let us now apply Gauss-Jordan elimination to the augmented matrix of the system:

$$\left[\begin{array}{ccccc|c} 2 & 6 & 2 & 2 & -1 & -7 \\ 2 & 6 & 1 & -1 & 2 & 6 \\ 3 & 9 & -1 & -9 & 1 & -6 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccccc|c} \boxed{1} & 3 & 0 & -2 & 0 & -3 \\ 0 & 0 & \boxed{1} & 3 & 0 & 2 \\ 0 & 0 & 0 & 0 & \boxed{1} & 5 \end{array}\right]$$

(I will leave you the joy of going through all the steps in Gauss-Jordan elimination above, as you did in Lecture 2). The matrix on the right is in reduced row echelon form (the pivots are boxed) and it corresponds to the system:

$$\begin{bmatrix} \boxed{1} & 3 & 0 & -2 & 0 \\ 0 & 0 & \boxed{1} & 3 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 5 \end{bmatrix} \tag{67}$$

By Fact 3, finding the general solution of the system (66) is equivalent to finding the solution of the system (67) (note: the right-hand sides of (66) and (67) are different, because the right-most column must also undergo Gauss-Jordan elimination together with the rest of the matrix) so let us solve the latter. To do so, let us look at the coefficient matrix:

$$R = \begin{bmatrix} \boxed{1} & 3 & 0 & -2 & 0 \\ 0 & 0 & \boxed{1} & 3 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

and identify the pivot columns (these are columns 1,3 and 5 in the example at hand) and the free columns (these are columns 2 and 4 in the example at hand). Therefore, the pivot variables are x_1, x_3, x_5 and the free variables are x_2, x_4. Now we make the following observation:

For any choice of free variables x_2, x_4 there exists a unique choice

of pivot variables x_1, x_3, x_5 which solve the system (67)

So we keep x_2, x_4 as arbitrary numbers, and solve for x_1, x_3, x_5 in (67) by back substitution:

$$\begin{cases} x_1 + 3x_2 - 2x_4 = -3 \\ x_3 + 3x_4 = 2 \\ x_5 = 5 \end{cases} \quad \rightsquigarrow \quad \begin{cases} x_1 = -3 - 3x_2 + 2x_4 \\ x_3 = 2 - 3x_4 \\ x_5 = 5 \end{cases}$$

So we conclude that the general solution to the system (67) (which is the same as the general solution to the system (66), by the very nature of Gauss-Jordan elimination) is:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}_{\text{general}} = \begin{bmatrix} -3 - 3a + 2b \\ a \\ 2 - 3b \\ b \\ 5 \end{bmatrix} \tag{68}$$

(we relabeled $x_2 = a$ and $x_4 = b$ to emphasize the fact that they are arbitrary real numbers). So what if you didn't need the general solution of (66) and you were satisfied with a particular solution? One easy way to get a particular solution is to just plug in $a = b = 0$, thus obtaining:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}_{\text{particular}} = \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 5 \end{bmatrix} \tag{69}$$

You could also have gotten this by setting all the free variables equal to 0 in (67) and solving for the pivot variables by using back substitution. Similarly, if you subtract the particular solution (69) from the general solution (68), you get:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}_{\text{general}} - \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}_{\text{particular}} = \begin{bmatrix} -3a+2b \\ a \\ -3b \\ b \\ 0 \end{bmatrix} \tag{70}$$

The right-hand side is just the general element of the nullspace of the matrix R.

Let us just mention one thing which could have gone wrong in the procedure above, and how to remedy it. It could have been that the reduced row echelon form of the coefficient matrix had a full row of zeroes. For example, suppose that instead of the system (67) we would have encountered:

$$\begin{bmatrix} \boxed{1} & 3 & 0 & -2 & 0 \\ 0 & 0 & \boxed{1} & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 5 \end{bmatrix} \tag{71}$$

(when doing elimination, it is customary to put all the zero rows at the very bottom of the matrix, which we can achieve by suitable row exchanges). It is easy to see that the system above has no solutions, because equating the third rows of equality (71) to each other would force us to have $0 = 5$, which is impossible. However, if the system were instead:

$$\begin{bmatrix} \boxed{1} & 3 & 0 & -2 & 0 \\ 0 & 0 & \boxed{1} & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 0 \end{bmatrix} \tag{72}$$

then we do have solutions. To find these solutions, just throw out the zero row at the bottom of the equality (72) and proceed as we did on the previous page. The upshot is that for any matrix R in reduced row echelon form:

a system $R\boldsymbol{v} = \boldsymbol{c}$ has solutions if and only if the vector
$\boldsymbol{c}$ has entry 0 corresponding to every full row of zeroes of R

Definition 7. *The **rank** r of a matrix A is equal to the number of its pivot columns.*

Here and throughout, the pivot columns of a matrix A refer to those columns where the pivots lie in the (reduced) row echelon form of A.

If the matrix A is $m \times n$, then the rank satifies the inequality:

$$\boxed{r \leq \min(m, n)}$$

since the number of pivots cannot surpass either the number of rows or of columns. We say that:

- A has full row rank if $r = m$. In this case, the column space of A is as big as possible, i.e. $C(A) = \mathbb{R}^m$, and the system $A\boldsymbol{v} = \boldsymbol{b}$ has <u>at least one</u> solution $\boldsymbol{v}$ for any $\boldsymbol{b}$.

- A has full column rank if $r = n$. In this case, the nullspace of A is as small as possible, i.e. $N(A) = 0$, and the system $A\boldsymbol{v} = \boldsymbol{b}$ has <u>at most one</u> solution $\boldsymbol{v}$ for any $\boldsymbol{b}$.